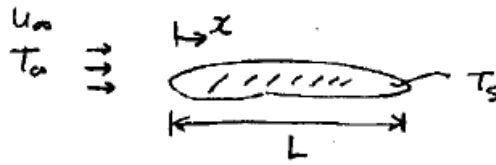


11. External flow heat convection



Q)

In fluid mechanics,

- Shear stress $\tau(x)$?
- Average shear stress $\bar{\tau}$?

In HT,

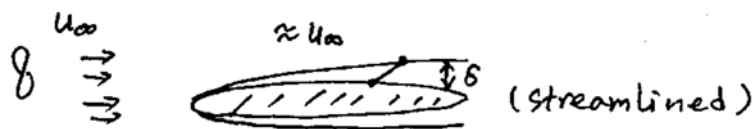
- Heat transfer coefficient $h(x)$?
- Average HT coefficient $\bar{h}$?

External flow vs. Internal flow: depending on the confinement of the boundary layer

- In external flow HT (HT between a solid surface and an (unrestrained) external flow, we need to know temperature distribution, i.e., $\frac{\partial T}{\partial y}\bigg|_{y=0} \therefore h = \frac{-k \partial T / \partial y|_{y=0}}{T_s - T_\infty}$, to determine h .
- In external flow analysis, we need to know velocity distribution, i.e., $\frac{\partial u}{\partial y}\bigg|_{y=0} \therefore \tau = \mu \frac{\partial u}{\partial y}\bigg|_{y=0}$, to determine τ .

$\Rightarrow$ Velocity & temperature field? Need to know boundary layer?

[Velocity (momentum) boundary layer]



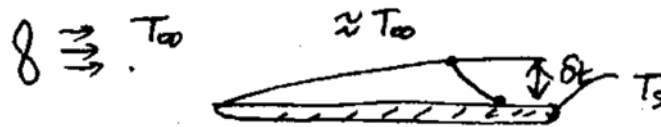
Q) Velocity field?

A) Flow field (velocity) almost unaffected except for a **thin** layer

- Velocity changes from 0 (no slip) to u_∞ across the layer δ , i.e., $u(\delta) \equiv 0.99u_\infty$
- Note BL is thin! For air $u_\infty = 10$ mph, $\delta = 9.5$ mm at $x = 1$ m (air $\nu = 1.589 \times 10^{-5}$ m²/s)
- Two flow regions:

- A thin fluid layer in which velocity gradients and shear stresses are large (BL region)
- An outer region in which velocity gradients and shear stresses are negligible (irrotational flow)
- Shear stress $\sim$ inversely proportional to BL thickness $\tau = \mu \frac{\partial u}{\partial y} \Big|_{y=0} \cong \mu \frac{\Delta u}{\Delta y} \sim \mu \frac{u_\infty}{\delta}$

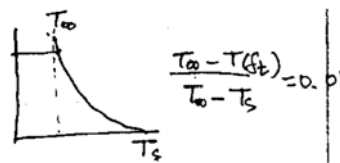
[Thermal boundary layer]



Q) Temperature field?

A) Temperature field almost unaffected except for a **thin** layer

- Temperature changes from T_s to T_∞ across the layer δ_t , i.e., $\frac{T_s - T(\delta_t)}{T_s - T_\infty} \cong 0.99$



- Thermal BL is thin, too! Typically same order of magnitude as velocity BL
- Two flow regions:
 - A thin fluid layer in which temperature gradient and heat transfer are large (thermal BL region)
 - An outer region in which temperature is uniform (isothermal flow, no HT)
- Heat flux (heat transfer coeff.) $\sim$ inversely proportional to thermal BL thickness

$$q'' = -k \frac{\partial T}{\partial y} \Big|_{y=0} \cong k \frac{\Delta T}{\Delta y} \sim k \frac{\Delta T}{\delta_t}$$

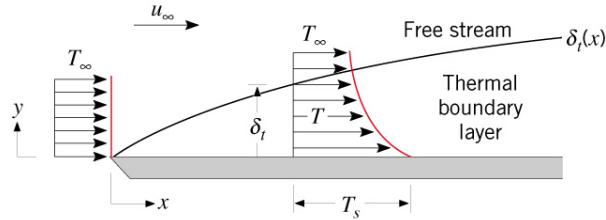
$$q'' = h \Delta T \rightarrow h \sim k \frac{\Delta T}{\delta_t}$$

In the BLs, $\frac{\partial}{\partial y} \gg \frac{\partial}{\partial x}$ (Note boundary layers are thin.)

$\Rightarrow$ Navier Stokes equations (momentum equations) and energy equation become simpler

[Boundary layer equations]

For 2-d, steady, laminar, incompressible flow over a flat plate with constant thermal properties



In the boundary layer

Governing equations

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2}$$

$$-\frac{1}{\rho} \frac{\partial p}{\partial y} = 0$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

Boundary conditions:

$$u(x, y=0) = v(x, 0) = 0$$

$$u(x, y=\infty) = u_\infty$$

$$T(x, 0) = T_s$$

$$T(x, \infty) = T_\infty$$

<Note>

- ① 4 unknowns (u, v, p, T) and 4 equations
- ② y -momentum equation means that pressure is uniform everywhere

$$\frac{\partial p}{\partial y} = 0 \Rightarrow p = f(x) \Rightarrow p = p_\infty(x) \Rightarrow p = p_\infty, \frac{\partial p}{\partial x} = 0$$

- ③ The above equations can be solved analytically using the similarity solution method (Blasius solution)